

Minimum-Entropy $\mathcal{H}_\infty$ Control of a Helicopter

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An explicitly parameterized simulation study

Abstract—This study implements minimum-entropy $\mathcal{H}_\infty$ control of a simulated helicopter. Its objective describes closed-loop disturbance response, not thermodynamic entropy. Numerical synthesis checks and seven paired nonlinear flight cases pass, with explicit PID landing support.

Index Terms—Helicopter, minimum entropy, robust control.

I. DEFINE THE CONTROL OBJECTIVE

For a stable, strictly proper closed-loop map $T : w \mapsto z$, disturbance and noise inputs w and regulated outputs z have declared weights. The Mustafa–Glover problem requires internal stability, $\|T\|_\infty < \gamma$, and minimizes [1], [2]

$$\begin{aligned} \mathcal{I}_\gamma(T) &= -\frac{\gamma^2}{2\pi} \int_{-\infty}^{\infty} \ln \det W(\omega) d\omega, \\ W(\omega) &= I - \gamma^{-2} T(j\omega)^* T(j\omega). \end{aligned} \quad (1)$$

The star denotes conjugate transpose. This frequency-domain criterion approaches $\|T\|_2^2$ as $\gamma \rightarrow \infty$. It is not heat generation in J/K and does not promise lower electrical consumption. Robustness to model uncertainty requires a specified uncertainty interconnection and bound.

II. SYNTHESIZE THE CENTRAL CONTROLLER

A nominal hover linearization and weighting filters define

$$\begin{aligned} \dot{x}_\ell &= A x_\ell + B_1 w + B_2 u, \\ z &= C_1 x_\ell + D_{12} u, \\ y &= C_2 x_\ell + D_{21} w. \end{aligned} \quad (2)$$

The design uses 20 local mechanical states, with three attitude-error coordinates; thermal states remain in the nonlinear plant. Three stable position filters, $\dot{\eta} = e_p - 0.04\eta$, extend the design to 23 states. Twelve kinematic sensors, six acceleration disturbances and twelve noise inputs define the normalized channels. All weights and matrices are published.

The normalized DGKF construction requires stabilizability, detectability, channel normalization, stabilizing nonnegative Riccati solutions X, Y , and $\rho(XY) < \gamma^2$ [3]. Setting the free controller parameter to zero gives the minimum-entropy realization at the chosen feasible γ .

The online law updates a dynamic state, $\dot{x}_K = A_K x_K + B_K y$, $u_\delta = C_K x_K$; the Riccati problems are solved before flight. A zero-order-hold realization updates at 100 Hz; sampled stability is checked separately.

III. DECLARE THE NONLINEAR BOUNDARY

The 2 kg uncalibrated plant retains nonlinear rotor, thermal and contact dynamics. MuJoCo advances the rigid body at 0.01 s; rotor states use RK4 with governed rotor speeds. Both controllers receive matching sensor noise and no true wind or mass in feedback. The simulator supplies a spooled, mass-specific initial trim; discovering it is not demonstrated. Pitch/slew limits, reference feedforward, anti-windup and PID landing support lie outside the linear theorem. Every hold and support step counts in the 70 s mission.

IV. VALIDATION AND PROJECT

The nominal design has $\gamma = 1.5134$ and a numerical bounded-real gain upper bound of 1.3389. The closed loop is stable; its sampled spectral radius is 0.99960008. The entropy is 17.5046 and the H_2 norm is 3.6367; independent Riccati checks, Lyapunov calculation, refined frequency integration, and a large- γ limit check agree.

All 15 trials pass: seven 70 s cases for each controller, plus a disabled-control check. Cases cover wind/gusts, $\pm 20\%$ mass and threefold sensor noise. No audited physical or landing limits are violated. $\mathcal{H}_\infty$ RMS error is 0.082–0.143 m, versus 0.139–0.239 m for the retuned PID. $\mathcal{H}_\infty$ supplies 93.04–93.16% of all commands; the rest are trim hold and PID landing support. No failure fallback occurs.

Limiters do intervene, especially under increased sensor noise. The linear certificate therefore does not prove the complete nonlinear flights robust. These paired results establish neither universal superiority nor hardware performance. Energy and thermal entropy are separate measurements. Code, assumed parameters, setup, viewer, contract, and evidence:

<https://github.com/savethebeesandseeds/Universalis-Materiae-Dexteritas>
Directory: `projects/helicopter`.

REFERENCES

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